Data Value Explorer

Scoring Model: draft for review

Formula version MVP-v2.0 · Source: MVP PRD v0.3, section 5 · Companion to the Scoring Model walkthrough in the explorer

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> **Status of this document.** This is a draft proposal. The weights, band edges, label thresholds and quadrant split are proposed defaults, accepted for the MVP as placeholders until calibration with real data.
> **Configurable scoring methods are planned for a future iteration.** All data in the explorer is synthetic, and the MVP generates the four dimension scores directly rather than computing them from measures. Sections 3 and 4 describe the measure-level design that real data collection will feed.

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1. Purpose and overview

The scoring model gives every data product one score per dimension each month, so the organisation can compare products, see change over time and decide what to keep, optimise, maintain or review for retirement. The unit of scoring is the **data product**: a logical set of data assets with one owner.

Scores are 0 to 100 and **higher is always better** in every dimension: a high Cost score means efficient, a high Risk score means safe (D06). Every score record stores the formula version used to compute it.

1. **Measures** are collected for each data asset, upstream object, data product or the whole warehouse.
2. Tag-based measures (PII level, privacy confirmed) are derived from the asset tag history.
3. Asset and upstream costs are **allocated** to data products (section 8).

4. Measures are **aggregated** to the data product with the rule in each measure definition.
5. Each measure is **normalised** to 0-100 (log band, linear or lookup).
6. **Criterion score** = weighted mean of its normalised measures (equal weights unless configured).
7. **Dimension score** = weighted sum of its criterion scores.
8. A **label** is mapped from each dimension score.
9. **Composite** = weighted sum of the four dimension scores; the risk flag and quadrant are derived.

2. The four dimensions

Dimension	Question it answers	Direction	Weight in composite	Labels (high / mid / low)
Value	Is the data product used, and does its owner say it matters?	High score = High value	34%	High / Medium / Low
Quality	Can people trust the data, and is it documented?	High score = High quality	33%	High / Medium / Low
Cost	What does it cost to keep, and how much of that is avoidable?	High score = Efficient (low cost)	23%	Efficient / Moderate / Expensive
Risk	What happens if the data is leaked, misused or wrong?	High score = Safe (low risk)	10%	Safe / Moderate / Exposed

3. Measures and criteria

Criterion weights sum to 100% within each dimension. Level: A = Data asset; U = Upstream object; P = Data product; W = CDW (whole warehouse). **Planned** means no data is collected for the measure yet.

Value (34% of composite)

Is the data product used, and does its owner say it matters? High score = High value.

Criterion (weight)	Measure	Lvl	Unit	Aggregation	Normalisation	Status
Analytics consumption (35%)	Power BI semantic models using assets	A	count	distinct count	log band	Planned
	Power BI reports using assets	A	count	distinct count	log band	Planned
	Monthly report views	A	count	sum	log band	Planned
	Direct distinct users	A	count	distinct count	log band	Planned
	Direct queries per month	A	count	sum	log band	Planned
	Consuming data products (reuse count)	P	count	distinct count of products with is_reuse = true	log band	Planned
ML/AI consumption (15%)	ML training jobs reading assets	A	count	distinct count	log band	Planned

Criterion (weight)	Measure	Lvl	Unit	Aggregation	Normalisation	Status
	AI agents / agentic workflows reading assets	A	count	distinct count	log band	Planned
	ML/AI read operations per month	A	count	sum	log band	Planned
Perceived value: decisions (25%)	Owner attestation: decisions	P	int, 1-5	n/a	lookup	Planned
Perceived value: operations (25%)	Owner attestation: operations	P	int, 1-5	n/a	lookup	Planned

Quality (33% of composite)

Can people trust the data, and is it documented? High score = High quality.

Criterion (weight)	Measure	Lvl	Unit	Aggregation	Normalisation	Status
Profiling and tests (30%)	Completeness	A	%	size-weighted mean	linear	Planned
	Validity	A	%	size-weighted mean	linear	Planned
	Quality test pass rate	A	%	tests-weighted mean	linear	Planned
Reliability (25%)	ADF ETL success rate	U	%	runs-weighted mean (rule R1)	linear	Planned
	On-time arrival rate	U	%	runs-weighted mean (rule R1)	linear	Planned
Metadata maturity (20%)	has_description, has_owner, has_steward, has_lineage, has_glossary_link	A	bool (5 monthly measures)	share of assets = true, then mean of 5	linear	Planned
Attestation: decisions (12.5%)	Owner attestation: decisions	P	int, 1-5	n/a	lookup	Planned
Attestation: operations (12.5%)	Owner attestation: operations	P	int, 1-5	n/a	lookup	Planned

Cost (23% of composite)

What does it cost to keep, and how much of that is avoidable? High score = Efficient (low cost).

Criterion (weight)	Measure	Lvl	Unit	Aggregation	Normalisation	Status
Storage (35%)	Allocated size, all layers (Raw Data, Curated, BIPRSD)	A	GB	allocated sum	log band, inverse	Planned
Compute (30%)	Stored procedure CPU time	U	CPU-hours	allocated sum	log band, inverse	Planned
	Stored procedure duration	U	hours	allocated sum	log band, inverse	Planned

Criterion (weight)	Measure	Lvl	Unit	Aggregation	Normalisation	Status
	ADF run cost	U	CAD	allocated sum	log band, inverse	Planned
Duplication (10%)	is_duplicate (asset overlaps or is a near-copy)	A	bool	share of assets = true (rule R2)	linear, inverse	Planned
Time tax (25%)	Hours for changes, reconciliation, incidents (3 measures)	P	hours	sum	log band, inverse	Planned

Risk (10% of composite)

What happens if the data is leaked, misused or wrong? High score = Safe (low risk).

Criterion (weight)	Measure	Lvl	Unit	Aggregation	Normalisation	Status
Privacy sensitivity (40%)	PII level and Privacy-confirmed, derived from tag history	A	string; bool	max of effective level (rule R3)	lookup (table L1)	Planned
Exposure (35%)	External reporting, ML use, automated decisions (3 flags)	A	bool	any	lookup (table L2)	Planned
	Access breadth (distinct principals)	A	count	distinct count	log band, inverse	Planned
Decision and reputation (25%)	Owner attestation of impact if wrong or leaked	P	int, 1-5	n/a	lookup, inverse	Planned

4. Turning measures into 0-100

- **Log bands** (counts, sizes, costs): seven bands whose edges roughly multiply, because these quantities span orders of magnitude. Edges are fixed absolute values, not percentiles (D11).
- **Linear** (percentages and shares): the value is the score; an inverse measure scores 100 minus the value.
- **Lookup** (owner attestations, privacy and exposure flags): a fixed table.

Score per band

Band	Inverse measures (cost, risk)	Non-inverse measures (value)
below e1	100	5
e1 to e2	85	20
e2 to e3	70	40
e3 to e4	55	55
e4 to e5	40	70
e5 to e6	20	85
e6 and above	5	100

Zero rule: a non-inverse count measure with value 0 scores 0. No consumption means no value from that measure.

Worked example: storage (inverse)

Bands are defined as a share of total CDW size at calibration, then frozen as absolute GB (synthetic calibration total: 20,000 GB).

Share of CDW	Frozen band (GB)	Storage score
< 0.1%	< 20	**100**
0.1% - 0.3%	20 - 60	**85**
0.3% - 1%	60 - 200	**70**
1% - 3%	200 - 600	**55**
3% - 10%	600 - 2,000	**40**
10% - 30%	2,000 - 6,000	**20**
>= 30%	>= 6,000	**5**

A product using 15% of the CDW (3,000 GB) falls in the 2,000 to 6,000 GB band and scores 20.

Band edges for every log-band measure (monthly, data product level)

Measure	Unit	Direction	e1	e2	e3	e4	e5	e6
Power BI semantic models	count	non-inverse	1	2	4	8	15	30
Power BI reports	count	non-inverse	1	3	10	25	60	150
Monthly report views	count	non-inverse	10	100	500	2,000	10,000	50,000
Direct distinct users	count	non-inverse	1	3	10	25	60	150
Direct queries	count	non-inverse	10	100	1,000	5,000	25,000	100,000
Consuming data products (reuse)	count	non-inverse	1	2	3	5	10	20
ML training jobs	count	non-inverse	1	2	3	5	10	20
AI agents / agentic workflows	count	non-inverse	1	2	3	5	10	20
ML/AI read operations	count	non-inverse	10	100	1,000	10,000	100,000	1,000,000
Allocated storage	GB	inverse	20	60	200	600	2,000	6,000
Stored procedure CPU time	CPU-hours	inverse	1	5	20	60	200	600
Stored procedure duration	hours	inverse	2	10	40	120	400	1,200
ADF run cost	CAD	inverse	10	50	200	600	2,000	6,000
Time tax (sum of 3 measures)	hours	inverse	2	5	10	20	40	80
Access breadth	distinct principals	inverse	5	15	40	100	250	600

A value x is in band k when $e(k-1) \leq x < e(k)$. Edges are calibrated to the synthetic data and will be recalibrated with real data (D22).

Owner attestation (1-5)

Owner's answer	Score	Score if inverse
1	0	100
2	25	75
3	50	50
4	75	25
5	100	0

5. Rolling up to the composite

```
Composite = 0.34 × Value + 0.33 × Quality + 0.23 × Cost + 0.1 × Risk
```

The defaults keep the ratio of the original 10/10/7/3 maximums. They are placeholders until calibration with real data (D21).

Rounding. Intermediate values (normalised measures, criterion scores) are stored with 2 decimals. Only dimension and composite scores are rounded to whole numbers (half up). Labels, the risk flag and the quadrant all use the rounded dimension scores.

6. Labels, risk flag and decision quadrant

Labels

Dimension	Score ≥ 67	Score 34 - 66	Score ≤ 33
Value	High	Medium	Low
Quality	High	Medium	Low
Cost	Efficient	Moderate	Expensive
Risk	Safe	Moderate	Exposed

Risk flag

A product whose Risk label is **Exposed** is flagged next to its composite everywhere in the explorer. The composite is not capped (D08).

Decision quadrant

X axis = Cost score (right = efficient), Y axis = Value score, split at 50 on both axes. Quality is point size and risk is point colour (D09).

Quadrant	Condition	Suggested action
Invest	Value ≥ 50, Cost ≥ 50	Keep and extend
Optimize	Value ≥ 50, Cost < 50	Reduce cost

Quadrant	Condition	Suggested action
Maintain	Value < 50, Cost ≥ 50	Monitor, promote use
Retire review	Value < 50, Cost < 50	Review for archive

7. Worked example

Illustrative numbers, not a product from the demo data.

Dimension	Score	Label	Weight	Score × weight
Value	**72**	High	34%	24.48
Quality	**58**	Medium	33%	19.14
Cost	**41**	Moderate	23%	9.43
Risk	**80**	Safe	10%	8.00

```
Composite = 61.05 → rounds to 61
```

Value 72 is ≥ 50 and Cost 41 is < 50, so the quadrant is **Optimize** (Reduce cost). Risk is Safe, so there is no risk flag.

8. Rules, cost allocation and design decisions

Calculation rules

- **R1 Reliability.** A data product uses all ADF pipelines that feed its assets. Rates are weighted by each pipeline's run count. Rates are not allocated: a shared pipeline counts in full for each product.
- **R2 Duplication.** Product measure = number of assets with is_duplicate = true divided by number of assets in the product.
- **R3 Privacy.** An asset's effective level is its PII level. If the PII level is Low or higher and the asset has no PrivacyReview.Confirmed tag, the effective level is one level higher (maximum High). The product level is the highest effective level of its assets.

Privacy lookup (L1) and exposure flags (L2)

Effective PII level	Privacy score
None	100
Low	70
Medium	40
High	0

Any asset in the product	Exposure flag score
No flag set	100
Deduct for external reporting	-40

Any asset in the product	Exposure flag score
Deduct for ML use	-30
Deduct for automated decisions	-30
Minimum	0

Exposure criterion = mean of the exposure flag score and the access breadth score.

Cost allocation

- Master data products (Employer, Worker, Provider) carry 100% of their asset and upstream costs.
- A consuming product that uses a master data asset records a reuse link (allocation factor 0). The reuse adds to the master data product's reuse count.
- Any other asset in more than one data product: cost divided equally (1/n).
- Upstream objects (stored procedures, ADF pipelines, Raw Data tables) that feed assets in more than one product: equal split.
- Control: allocated storage for all products plus unassigned CDW storage equals total CDW storage, per period, within 0.01 GB.

Design decisions behind the numbers

- **D06** Scores are 0 to 100. Higher is better in every dimension (high cost score = efficient; high risk score = safe).
- **D07** Composite score = weighted sum of dimension scores, with explicit, versioned weights.
- **D08** Risk gate: a product with risk label Exposed is flagged. The composite is not capped.
- **D11** Quantitative measures use fixed log-scale bands, frozen as absolute units at calibration time. No percentiles.
- **D21** Default weights, label thresholds and quadrant split are accepted for the MVP. They are placeholders until calibration with real data.
- **D22** Log-band reference values are absolute monthly values at data product level, set for the synthetic data. They are recalibrated with real data in the target solution.

9. Questions to expect

Why these weights (34 / 33 / 23 / 10)?
They are proposed defaults that keep the ratio of the original 10/10/7/3 maximums (D21). They are placeholders until calibration with real data, and the weights are explicit and versioned so any change is traceable.

Why log-scale bands instead of a straight scale or percentiles?
Usage, size and cost span orders of magnitude, so equal-width bands would put almost every product in the bottom band. Bands are frozen as absolute units rather than percentiles (D11), so a product's score does not change just because other products changed.

Isn't owner attestation subjective?
Yes. It is used where no system can measure the answer (perceived value, impact if wrong or leaked), on a 1-5 scale with a fixed lookup, and it is only part of each dimension. Value and Quality use 50% and 25% attestation respectively. The roadmap includes a plan for collecting attestations from data owners.

A product with a terrible Risk score can still have a high composite. Why?
Risk carries 10% of the composite on purpose, and the composite is not capped (D08). Instead, any product whose Risk label is Exposed is flagged wherever the composite is shown, so the risk cannot be hidden by good scores elsewhere.

Why is Cost (and not Risk or Quality) one of the two quadrant axes?
The quadrant answers a portfolio question: is this worth what it costs? Value against Cost gives four actions (keep and extend, reduce cost, monitor and promote use, review for archive). Quality is shown as point size and risk as point colour (D09).

Are these numbers real?
No. All data in this explorer is synthetic, and this MVP generates dimension scores directly instead of computing them from measures. The measure-level model on these pages is the design that real data collection will feed.

10. Limitations and what's next

> **Limitations of this MVP.** All data is synthetic. Dimension scores are generated directly, not computed from measures. Weights, band edges and thresholds are uncalibrated placeholders and cannot yet be changed without editing the configuration file.

Next steps

1. Review the functionality of the Data Value Explorer.
2. Implement the ability to capture user feedback.
3. Review and refine the scoring model.
4. Implement configurable scoring methods.
5. Define the measures to be collected and their sources.
6. Develop a roadmap for the collection of measures and data owner attestation over the data products.

> **Configurable scoring methods.** A future iteration will make the weights, bands, thresholds and quadrant split configurable instead of fixed defaults.